Recall: $f_{\mu^{(n)}}(x) = \frac{1}{n}\sum a_j \sigma(w_j \cdot x) = \int \varphi(\theta, x)\,\mu(d\theta)$

$\theta := (a, w) \in \Theta \subseteq \mathbb{R}^{d+2}$ (weights)

$\varphi(\theta, x) = a\,\sigma(w \cdot x)$

$$\Rightarrow \partial_t \mu_t = -\operatorname{div}\left(\mu_t\left(-D_\theta \int \varphi(\theta,x)\left(f_{\mu_t}(x) - f_*(x)\right)\mathbb{P}(dx)\right)\right) + \frac{\eta}{2}\Delta\mu_t$$

$$= -\operatorname{div}\Bigg(\mu_t\Bigg(-D_\theta\Bigg(\underbrace{\int \varphi(\theta,x) f_*(x)\,\mathbb{P}(dx)}_{V(\theta)}$$

$$- \int \underbrace{\int \varphi(\theta,x)\varphi(\theta',x)\,\mathbb{P}(dx)}_{U(\theta,\theta')}\,\mu_t(d\theta')\Bigg)\Bigg)\Bigg) + \frac{\eta}{2}\Delta\mu_t$$

$$= -\operatorname{div}\Bigg(\mu_t\Bigg(-D_\theta\underbrace{\left(V(\theta) - \int U(\theta,\theta')\,\mu_t(d\theta')\right)}_{\Psi(\theta,\mu)}\Bigg)\Bigg) + \underbrace{\frac{\eta}{2}\Delta\mu_t}_{\operatorname{div}\,\eta\,\mu_t D_\theta \log\mu_t}$$

$$= -\operatorname{div}\left(\mu_t\left(-D_\theta\left(\Psi(\theta,\mu) + \eta\log\mu\right)\right)\right).$$

Thm: $\mu_t \longrightarrow \mu_*$ where $\mu_* \simeq \exp\left(-\frac{\Psi(\theta,\mu_*)}{\eta}\right)$

Proof: We can write $\mu_t = -\operatorname{div}\left(\mu_t\left(-D_\vartheta\left(\frac{\delta}{\delta\mu}R'(\mu_t)\right)\right)\right)$

for $R'(\mu_t) = \iint 2V(\vartheta) + \int U(\vartheta,\vartheta')\mu(d\vartheta)\mu(d\vartheta) + \eta\int \mu_t \log \mu_t$

Fact: $\frac{d}{dt}R'(\mu_t) \leq 0$, $R'(\mu) \geq C \implies \lim_{t\to\infty} R(\mu_t)\ \exists$

$$\implies \lim_{t\to\infty}\int\left(-D_\vartheta\left(\Psi(\vartheta,\mu) + \eta\log\mu\right)\right)^2\mu_t = 0$$

unif. tight $\implies \exists\, \mu_* \text{ s.t. } \mu_{t_k} \xrightarrow{k\to\infty} \mu_*$

$\implies \mu_* = Z^{-1}\exp\left(-\frac{1}{\eta}\Psi(\vartheta,\mu_*)\right)$

. the above has a unique solution $\implies \mu_*$ is unique.

Prop: let $\vartheta_0^{(i)} \overset{iid}{\sim} \mu_0$ then $\forall t > 0$ $\mu_t^{(n)} \longrightarrow \mu_t$

Proof: Consider $\frac{d}{dt}\bar\vartheta_t^{(i)} = \underbrace{-D_\vartheta\left(-\int U(\bar\vartheta_t,\vartheta)\mu_t(d\vartheta) + V(\bar\vartheta_t)\right)}_{\bar F(\vartheta)}$

$$\frac{d}{dt}\vartheta_t^{(i)} = \underbrace{-D_\vartheta\left(-\int U(\vartheta_t^{(i)},\vartheta)\mu_t^{(n)}(d\vartheta) + V(\vartheta_t^{(i)})\right)}_{F^{(n)}(\vartheta)}$$

Fact: if $\bar\vartheta_0^{(i)} = \vartheta_0^{(i)} \overset{iid}{\sim} \mu_0$ then $\mu_t^{(n)}$ obeys MFPDE

$$\|\vartheta_t^{(i)} - \bar\vartheta_t^{(i)}\| = \left\|\int_0^t F^{(n)}(\vartheta_s^{(i)}) - \bar F(\bar\vartheta_s^{(i)})\,ds\right\|$$

$$\leq \int_0^t \underbrace{\| F^{(n)}(\vartheta_s^{(j)}) - \bar{F}(\vartheta_s^{(j)}) \|}_{\leq \frac{K}{N} \sum_{l=1}^{N} \| \vartheta_s^{(l)} - \bar{\vartheta}_s^{(l)} \| + \frac{K}{N}} ds$$

$$+ \int \underbrace{\| \bar{F}(\vartheta_s^{(j)}) - \bar{F}(\bar{\vartheta}_s^{(j)}) \|}_{\leq K^2 \| \bar{\vartheta}_s^{(j)} - \vartheta_s^{(j)} \|} ds$$

$$\Longrightarrow \max_j \sup_{t \in (0,T)} \| \vartheta_t^{(j)} - \bar{\vartheta}_t^{(j)} \| = \Delta_t(N)$$

$$\Delta_t(n) \leq K \int_0^t \Delta_s(n)\, ds + \frac{Kt}{n}$$

$$\Longrightarrow \Delta_t(n) \leq \frac{K}{n} e^{Kt} \xrightarrow[n \to \infty]{} 0$$

and claim follows from

$$W_2(\mu_t^{(n)}, \mu_t) \leq \mathbb{E}(\Delta_t(n))$$

A note on generalization

Consider $f_* = \sigma_*(w_* \cdot x)$

Know: if σ_* is not a polynomial assuming that $d^{\ell} \leq n \leq d^{\ell+1}$ we have

$$R^{NTK}(\theta^*) \geq b_\ell > 0$$

However: $\exists$ a single layer NN achieving a generalization